

Evidence-First Bengali Hallucination Detection: A Deterministic Selective Cascade

Touhidul Alam Seyam¹ MD. Abtahee Kabir² Joyeta Barua Moni²
Noore Tamanna Orny²

¹BGC Trust University Bangladesh ²Chittagong University of Engineering & Technology
Team Huntrix, IUT 12th ICT FEST Datathon, 2026

Abstract

We present a selective cascade for detecting hallucinations in Bengali model responses. Instead of asking one model to judge every row, the system first applies exact source-lineage checks, task-specific solvers, context grounding, and relation-typed evidence. A quantized open-weight language model judges only unresolved cases. This design raised public macro-F1 from 0.657 to 0.967 and obtained private macro-F1 0.982, ranking first in Phase 1. On 299 released labels, the full cascade reaches 0.9933 macro-F1; deterministic stages cover 225 rows without error. Three offline Kaggle runs on two Tesla T4 GPUs reproduced the scored 2,516-row output byte for byte in 440.87–593.43 seconds. A 5,000-row same-mix run took 14.60 minutes, while an all-uncertain stress estimate remained below 5.5 hours. The study shows that careful evidence admission, explicit abstention, and deterministic scheduling can outperform uniform LLM judging while satisfying a nine-hour offline inference limit.

1 Introduction

The Bengali LLM Hallucination Detection Challenge asks whether a candidate response is fabricated, unsupported, or factually wrong (label=0), or faithful (label=1). A row contains a Bengali prompt, a response, and sometimes a supporting context. The released development set has 299 labels; Phase 1 contains 2,516 unlabeled rows. The held-out Phase 2 set contains approximately 5,000 rows from a wider range of sources (Ruhan et al., 2026; Competition Organizers, 2026).

This is not one homogeneous classification problem. The benchmark mixes context entailment, arithmetic, multiple-choice questions, morphology, idioms, legislation, history, literature, and civic information. It also inherits task diversity from Bengali hallucination research such as BenHalluEval (Adib et al., 2026). A generic LLM

prompt must solve all of these relations with one evidence policy, and our early judges were both prompt-sensitive and expensive.

Our central decision was therefore to make the classifier selective. A deterministic component may label a row only when its scope, evidence source, and semantic relation are explicit. Otherwise it abstains and sends the row to a bounded LLM judge. This separation gives exact tasks to exact methods and reserves model capacity for genuine ambiguity. The final system has three goals: high recall for the hallucinated class, protection against semantically related but irrelevant retrieval, and reproducible offline execution on Kaggle T4×2.

2 Method

Figure 1 shows the frozen v91 cascade. For input row x , the router returns either a label $d(x) \in \{0, 1\}$ with provenance or an abstention symbol $\perp$. The final prediction is

$$\hat{y}(x) = \begin{cases} d(x), & d(x) \neq \perp, \\ J(x), & d(x) = \perp, \end{cases}$$

where J is a fixed-order, grammar-constrained open-weight judge.

2.1 Normalization and Routing

The first stage normalizes Bengali and Latin digits, Unicode punctuation, spacing, answer aliases, and inline answer options. It also distinguishes absent context, evidential context, and instructional text. The last distinction matters because a grammar instruction can mention the requested concept without providing the answer.

Exact lineage checks match public benchmark or question-answer records only when normalized prompt, response, and applicable context fields agree. Reliability-weighted answer banks can resolve a row when the accepted answer relation is

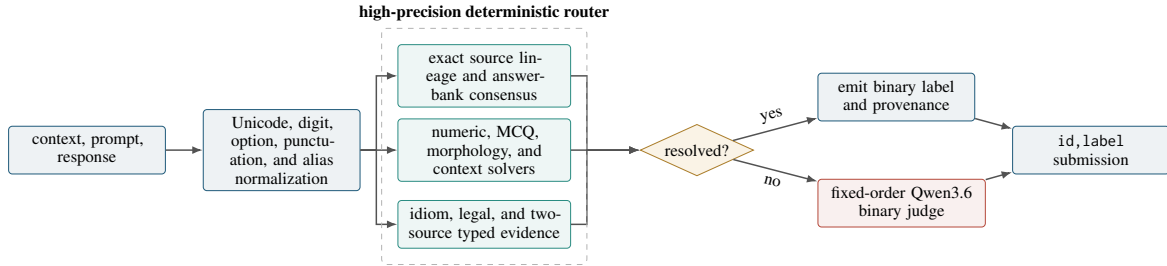


Figure 1: Frozen v91 inference path. Deterministic stages emit a label only for a relation-matched case; unresolved rows reach the open-weight judge.

unambiguous. Approximate retrieval may nominate a candidate, but cannot overrule an exact source or an intrinsic context contradiction.

2.2 Structured and Context Solvers

Small solvers cover arithmetic, numeric ranges, Bengali compound number words, multiple-choice serialization, morphology, compounds, and prefixes. Relation guards bind dates or numbers to the clause containing the requested entity and predicate. This prevents a valid year elsewhere in a passage from answering the wrong birth, death, publication, or establishment question.

For grounded rows, response claims are checked against the supplied passage. Exact answer spans are accepted only inside a semicolon-aware clause that also contains prompt anchors. Named entities, dates, negation, and alternatives are checked before external facts may contribute. Supplied context takes precedence over outside plausibility.

2.3 Evidence Admission and Conflict Resolution

Table 1 summarizes the admission policy. Idiom records distinguish figurative meaning from literal gloss, a known weakness in Bengali LLMs (Sakhawat et al., 2026). Legal records require the named act and section. A free-form fact requires two independent public sources and a typed relation. These guards were added after broad retrieval repeatedly found relevant passages that answered a neighboring question.

Every proposal records source family, relation, and confidence. Intrinsic context contradiction has highest priority. Exact authoritative lineage follows, then relation-aligned parsing and consensus evidence. If sources describe different relations, neither decides the row. If they claim the same relation but disagree, the system requires a stronger

Table 1: Evidence admission policy.

Evidence family	Requirement for emitting a label
Exact lineage	Prompt, response, and applicable context agree
Answer banks	Relation match and reliability-weighted consensus
Structured rule	Parsed values, entity, and predicate align
Context evidence	Same clause and prompt anchors; no contradiction
Idiom or law	Exact headword/section and requested sense/provision
Free-form fact	Typed relation and two independent sources
Fuzzy retrieval	Candidate generation only

exact source or abstains. We accepted a larger judge queue when that was safer than allowing weak retrieval to override a trusted decision.

2.4 Selective Open-Weight Judge

The final Phase 1 router leaves 158 rows unresolved. They are sorted by row ID and evaluated in fixed batches. Context rows receive one entailment view. No-context rows receive up to three short views: factual verification, strict exam grading, and a zero-bias check. A faithful decision requires unanimous positive votes; evaluation stops after the first zero.

The judge is Qwen3.6-27B Q4_K_P, a 16.332 GiB GGUF checkpoint, served by pinned llama.cpp b9637 across two T4 GPUs (Qwen Team, 2026; Gerganov and contributors, 2026). Thinking is disabled, and a grammar restricts generation to one binary token. Seeds, row order, prompt order, and batch boundaries are fixed.

3 Experiments

3.1 Protocol

We used the 299 released labels for validation and error analysis, not as proof that a large fine-tune would generalize. We tracked macro-F1, per-

Table 2: Representative scored iterations.

Version	Public F1	Main change
Baseline	0.657	character and embedding models
Gemma	0.742	12B QAT generic judge
Qwen	0.797	27B factual judge
v6	0.903	source and structured routing
v23	0.945	provenance repair
v37	0.961	source-lineage audit
v60	0.963	conservative fact corrections
v85	0.966	reproducible two-source anchor
v87	0.965	plausible edits; reverted
v91	0.967	typed idiom and relation guards

class F1, accuracy, deterministic coverage, and context/no-context behavior. A candidate rule entered the system only when it expressed a reusable relation and passed the regression suite. The public leaderboard represented roughly 30% of Phase 1, so a rounded score change was treated as aggregate evidence rather than a row label.

3.2 Iteration and Error Analysis

Figure 2 and Table 2 summarize the experiment path. Character and embedding models established a 0.657 baseline. Gemma 12B QAT and the Qwen judge improved factual reasoning but remained prompt-sensitive. v6 introduced source and structured routing and jumped to 0.903. Later gains came from lineage audits, stricter provenance, legal evidence, and relation guards rather than a larger model.

Several plausible approaches failed. A prompt-vote ensemble reached 0.8512 locally but only 0.783 publicly, exposing prompt overfitting. A TF-IDF verifier improved from 0.7879 to 0.8261 out of fold, yet remained weak on no-context cultural facts. Raw external-QA retrieval often found adjacent passages with the wrong relation or authority. A global substring rule confused instructional grammar context with evidence. Finally, v87 made two externally plausible fact edits and regressed from 0.966 to 0.965. We reverted it and tightened evidence admission.

This sequence changed how experiments were selected. Retrieval became proposal-only, context rules became relation-aware, and researched facts required both provenance and a typed predicate. Candidate batches without a defensible expected direction were quarantined rather than submitted.

3.3 Results

Table 3 separates released-label validation from leaderboard results. The full cascade makes two

Table 3: Validation and Phase 1 results.

Evaluation	Macro-F1	Other result
Released set (299)	0.993264	accuracy 0.993311
Deterministic (225)	1.000000	label-0 F1 1.000000
Phase 1 public	0.967	rank 5
Phase 1 private	0.982	rank 1

Table 4: Frozen deployment measurements.

Property	Measurement
Hardware	Kaggle Tesla T4×2; offline
Model	Qwen3.6-27B Q4_K_P; 16.332 GiB
2,516-row runs	440.87, 527.03, 593.43 seconds
5,000 same-mix run	876.14 seconds (14.60 minutes)
All-uncertain estimate	5.2–5.5 hours
Output	1,181 zeros; 1,335 ones
Regression suite	249 tests passing at freeze

errors on 299 released labels; both concern internally inconsistent controls, which we did not patch with row-specific exceptions. The deterministic subset covers 225 rows with no error. On Phase 1, v91 finished public rank 5 at 0.967 and private rank 1 at 0.982. The private improvement suggests that the conservative policy transferred beyond the public slice, but it does not prove that each of the five v91 changes generalized individually.

4 Reproducibility and Phase 2

4.1 Frozen Inference and Determinism

The notebook reads the organizer’s literal path:

```
/kaggle/input/competitions/
bengali-hallucination/test set.csv
```

It does not search for alternate filenames, branch on row count, call an external API, or load a saved Phase 1 prediction vector. The organizer can replace the file mounted at that path without editing a variable.

Reproduction requires more than a seed. The notebook sorts unresolved IDs, fixes batch boundaries, pins both llama.cpp server commands, waits for health checks, and assigns each view to the same server. The CSV writer preserves incoming IDs and a fixed id, label column order. Three full runs produced byte-identical files matching scored submission 54836339.

4.2 External Resources

The evidence inventory combines public Bengali resources rather than treating any one corpus as universal truth. QA lineage came from TyDi

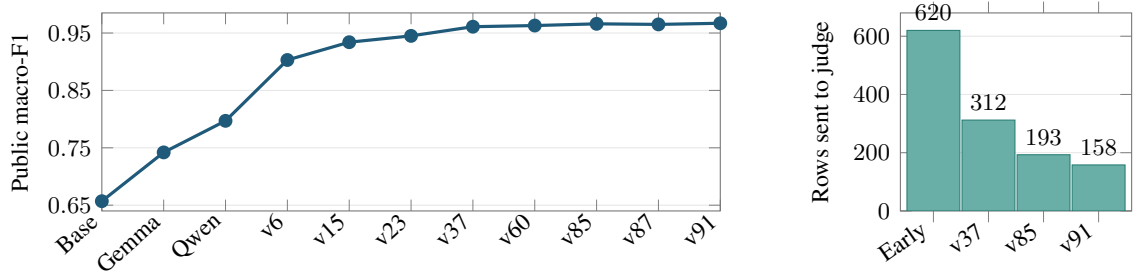


Figure 2: Left: public leaderboard progression. Right: deterministic routing reduced the Phase 1 judge queue by 74.5%. The largest accuracy gain followed the move from uniform judging to source and task routing.

QA, Suman Bengali QA, BUFFET, SQuAD-BN, BanglaRQA, IndicQA, Aya Indic, and public Bengali QA collections (Clark et al., 2020; Mondal, 2026; BUFFET Authors, 2023; CSE BUET NLP, 2026; Ekram, 2026; AI4Bharat, 2026; Cognitive Lab, 2026; Rasheduzzaman, 2026). Domain evidence included IndicXNLI, Bengali Wikipedia, textbooks, newspapers, Bangladesh legal corpora, history and literature QA, and public exam banks (Divyanshu, 2026; Wikimedia Foundation, 2026; Nishat, 2026; Nabil, 2026; Sakhawat, 2026b; Saha, 2026; Ramim, 2026; Lavlu, 2026; Badhon, 2026; Hishab, 2026; Wasi, 2026a,b). Bengali idiom and lexical records came from Bangla Bagdhara, BLUCK, BengaliDictionary, and a word-analogy collection (Sakhawat, 2026a; Abedin, 2026; Kamal, 2026; Mousumi, 2023). Wikidata supplied only corroborating structured facts (Wikidata Contributors, 2026). Sources with unclear redistribution terms were used for audit or retrieval, not copied into the public package.

4.3 Held-Out Domain Shift

The 5,000-row run duplicates the current source mixture and therefore measures throughput, not domain accuracy. The organizer preview instead names Common Crawl, Bengali Wikipedia, Banglapedia, government service pages, recent newspapers, legislation, NCTB textbooks, and Bengali literature (Competition Organizers, 2026). Exact Phase 1 lineage should cover less of this distribution. Table 5 states the fallback policy.

Unfamiliar sources bypass lineage and reach context reasoning or the judge; approximate matches are never forced. Runtime rises as deterministic coverage falls, but the all-uncertain estimate remains below nine hours.

No fine-tuned checkpoint is required. The package pins the public Qwen quantization, CUDA runtime, evidence dataset, and inference notebook.

Table 5: Routing policy for announced Phase 2 sources.

Source family	Primary route; abstention trigger
Common Crawl/news	Context check or judge; unsupported outside claim
Wikipedia/Banglapedia	Exact entity-relation match; alias ambiguity
Government services	Clause, date, fee, and office checks; stale page
Legislation	Exact act and section; conflicting or absent provision
NCTB/literature	Structured solver or exact work relation; edition mismatch
Unknown	Skip lineage and invoke the fixed judge

The router contains short, citation-backed facts from public sources, but no hidden labels or manually assigned row-label vector. Every output is recomputed from the input row and frozen assets.

5 Conclusion

The largest gain came from changing the unit of reasoning, not enlarging the model. Exact provenance and typed solvers handled cases with clear relations; the fixed judge handled the uncertain tail. This yielded a first-place private Phase 1 score, reproducible output, and substantial headroom under the Phase 2 runtime limit. The system also exposes why each deterministic decision was made and when it chose to abstain.

Limitations

The private score does not guarantee equal accuracy on the broader Phase 2 source mix. Exact lineage cannot replace general reasoning on unfamiliar civic, legal, textbook, and news content, and the 299 released labels are too small for reliable per-domain estimates. The all-uncertain timing is a projection, not an adversarial run. Evidence can age, conflict, or carry licensing constraints; high-stakes civic use still requires human escalation.

References

- Minhajul Abedin. 2026. [BLUCK: Bengali linguistic understanding and cultural knowledge](#). GitHub repository.
- Shefayat E Shams Adib, Ahmed Alfey Sani, Ekramul Alam Esham, Ajwad Abrar, Ishmam Tashdeed, and Md Taukir Azam Chowdhury. 2026. [Ben-HalluEval: A multi-task hallucination evaluation framework for large language models on bengali](#). *Preprint*, arXiv:2605.31483.
- AI4Bharat. 2026. [Indicqa](#). Hugging Face dataset.
- Badhon. 2026. [Bangla literature QA dataset](#). Hugging Face dataset.
- BUFFET Authors. 2023. [BUFFET: A benchmark for few-shot cross-lingual transfer](#). Hugging Face dataset.
- Jonathan H. Clark, Eunsol Choi, Michael Collins, Dan Garrette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki. 2020. [TyDi QA: A benchmark for information-seeking question answering in typologically diverse languages](#). *Transactions of the Association for Computational Linguistics*, 8:454–470.
- Cognitive Lab. 2026. [Aya dataset indic](#). Hugging Face dataset.
- Competition Organizers. 2026. [Phase 2 reveal and a note on data resources](#). Kaggle Discussion.
- CSE BUET NLP. 2026. [SQuAD-BN](#). Hugging Face dataset.
- Divyanshu. 2026. [Indicxnli](#). Hugging Face dataset.
- Sartaj Ekram. 2026. [Banglarqa](#). Hugging Face dataset.
- Georgi Gerganov and contributors. 2026. [llama.cpp: LLM inference in C/C++](#). GitHub repository, pinned release b9637.
- Hishab. 2026. [Bangla MMLU](#). Hugging Face dataset.
- Minhas Kamal. 2026. [Bengalidictionary](#). GitHub repository.
- Lavlu. 2026. [Bonolota bengali QA](#). Hugging Face dataset.
- Suman Mondal. 2026. [Bengali QA](#). Hugging Face dataset.
- Mousumi. 2023. [Word analogy bangla](#). GitHub repository.
- Zabir Nabil. 2026. [Bangla newspaper dataset](#). Hugging Face dataset.
- Md Nishat. 2026. [Bangla textbook](#). Hugging Face dataset.
- Qwen Team. 2026. [Qwen3.6-27B-FP8 model card](#). Hugging Face.
- Sabbir Ramim. 2026. [Bengali question answering: Pre-independence 1947–1971](#). Hugging Face dataset.
- Rasheduzzaman. 2026. [Bangla question answer pair 70k dataset](#). Hugging Face dataset.
- Ahmed Shafin Ruhan, Arefin Mizan, Jubaer, Shefayat, and Sushmit. 2026. [Bengali LLM hallucination detection challenge](#). Kaggle.
- Bipin Saha. 2026. [Bangla law question answering dataset](#). Hugging Face dataset.
- Adib Sakhawat. 2026a. [Bangla bagdhara](#). Kaggle dataset.
- Adib Sakhawat. 2026b. [Bangladesh legal acts dataset](#). Hugging Face dataset.
- Adib Sakhawat, Shamim Ara Parveen, Md Ruhul Amin, Shamim Al Mahmud, Md Saiful Islam, and Tahera Khatun. 2026. [When words don't mean what they say: Figurative understanding in bengali idioms](#). *Preprint*, arXiv:2602.12921.
- Azminetoushik Wasi. 2026a. [Bangla BCS questions](#). Hugging Face dataset.
- Azminetoushik Wasi. 2026b. [BCS 10th–40th GK/ICT questions](#). Hugging Face dataset.
- Wikidata Contributors. 2026. [Wikidata query service](#).
- Wikimedia Foundation. 2026. [Bengali wikipedia](#).